

# Lecture 20 Worksheet - Capstone Design Review

## Exercise A. Workload and Target Assumptions

| Item                   | Decision | Reason |
|------------------------|----------|--------|
| Target workload        |          |        |
| Input shape/dtype      |          |        |
| Native tile            |          |        |
| SRAM capacity          |          |        |
| Supported quantization |          |        |
| Fallback target        |          |        |

## Exercise B. Legal Op Set

| Op             | Legal Condition | Lowering | Fallback/Diagnostic |
|----------------|-----------------|----------|---------------------|
| matmul         |                 |          |                     |
| bias add       |                 |          |                     |
| activation     |                 |          |                     |
| requant        |                 |          |                     |
| unsupported op |                 |          |                     |

## Exercise C. Pass Pipeline Draft

| #  | Pass | Input -> Output IR | Invariant / Test |
|----|------|--------------------|------------------|
| 1  |      |                    |                  |
| 2  |      |                    |                  |
| 3  |      |                    |                  |
| 4  |      |                    |                  |
| 5  |      |                    |                  |
| 6  |      |                    |                  |
| 7  |      |                    |                  |
| 8  |      |                    |                  |
| 9  |      |                    |                  |
| 10 |      |                    |                  |
| 11 |      |                    |                  |
| 12 |      |                    |                  |

## Exercise D. Schedule and Memory Plan

| Buffer/Token      | Memory Space | Shape/DType | Lifetime / Policy |
|-------------------|--------------|-------------|-------------------|
| A tile            |              |             |                   |
| B tile            |              |             |                   |
| Accumulator       |              |             |                   |
| Output tile       |              |             |                   |
| DMA token/barrier |              |             |                   |

## Exercise E. Custom Dialect Contract

| Op         | Operands | Results | Attributes | Verifier Rules |
|------------|----------|---------|------------|----------------|
| npu.matmul |          |         |            |                |
| npu.dma    |          |         |            |                |

|             |  |  |  |  |
|-------------|--|--|--|--|
| npu.barrier |  |  |  |  |
| npu.requant |  |  |  |  |

#### Exercise F. Runtime ABI

| Area                     | Design | Test |
|--------------------------|--------|------|
| Tensor descriptor fields |        |      |
| Command buffer header    |        |      |
| Packet types             |        |      |
| Version/capability query |        |      |
| Error/fallback policy    |        |      |

#### Exercise G. Verification Plan

| Test               | Case | Expected Result |
|--------------------|------|-----------------|
| Positive FileCheck |      |                 |
| Negative verifier  |      |                 |
| Numerical golden   |      |                 |
| Cost model         |      |                 |
| Demo script        |      |                 |

#### Submission Checklist

- ☐ Design spec complete
- ☐ Pipeline spec included
- ☐ Pseudo MLIR included
- ☐ Runtime ABI included
- ☐ Tests included
- ☐ Cost model output included
- ☐ 10-minute presentation prepared